Pneumonia Detection using a Multi-scale Transformer Approach

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ABSTRACT

This project presents various improvement approaches to a framework for enhancing the detection of pneumonia using advanced computer vision techniques. Building on a foundational model, the project integrates ResNet and Vision Transformer (ViT) backbones, coupled with transformer modules, to optimize performance across multiple approaches. The framework incorporates dynamic weight fusion and position embeddings to enhance feature extraction and decision-making processes. Comprehensive experiments were conducted using pediatric pneumonia chest X-ray datasets, and the results demonstrate competitive accuracy and robustness in classification. The study highlights the benefits of combining traditional convolutional backbones with transformer-based architectures, offering improved interpretability and performance metrics, such as F1-score and AUC.

KEYWORDS

Machine Learning, Transformers, Pneumonia Detection

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1 INTRODUCTION

1.1 Problem Definition

Our project aims to efficiently and accurately detect Pneumonia from chest x-rays. We first segment the lung region from the x-rays and then detect Pneumonia from the segmented images. We are using the "Chest X-ray Dataset for Tuberculosis Segmentation" [2] to train and "Lung segmentation from Chest X-Ray dataset" [4] to test the segmentation model and Kermany [1] for classification. We selected a base model to work with and attempted various architectural changes to improve it's performance.

1.2 Literature Review

This project uses the work "Efficient and Accurate Pneumonia Detection Using a Novel Multi-Scale Transformer Approach" as it's base paper [5]. The authors of this paper have used a TransUnet model that integrates their specialized transformer module, which has fewer parameters compared to common transformers while maintaining performance. For classification, they employ pre-trained resnet models (ResNet-50 and ResNet-101) to extract multi-scale feature maps, which are then processed through their modified transformer module. The segmentation model is trained on the "Chest Xray Masks and Labels" dataset[3]. It is then employed on the Kermany[1] and Cohen datasets to isolate lung regions, enhancing subsequent classification tasks. Their approach achieves 92.79% and 95.11% accuracy in the Kermany and Cohen datasets respectively.

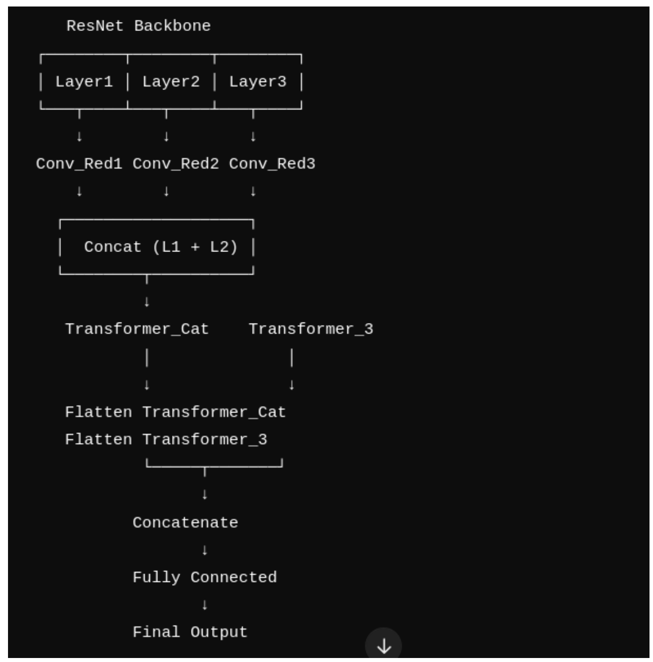

Figure 1: Visual Overview of the Base Paper Architecture

1.3 Gap

Despite advancements in deep learning for pneumonia detection, several critical and domain-specific research gaps persist:

- **Limited Generalization Across Diverse Demographics**: Existing models are often trained on datasets that lack diversity in terms of age groups, ethnicities, and clinical settings. This hinders the ability of these models to generalize effectively across different patient populations.
- **Insufficient Robustness to Imaging Variations**: Chest X-ray images can vary significantly due to differences in equipment, imaging protocols, and noise levels (e.g., artifacts from medical devices). Current models often struggle to maintain performance across datasets with such variations.

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- **Focus on Binary Classification**: Many studies approach pneumonia detection as a binary classification problem (e.g., pneumonia vs. normal). However, distinguishing between bacterial, viral, and other types of pneumonia is crucial for targeted clinical interventions, and this multi-class classification problem remains underexplored.
- **Suboptimal Feature Fusion Strategies**: Combining hierarchical features from convolutional models with global context from transformers is promising but remains under-optimized. Effective fusion mechanisms that leverage the complementary strengths of these architectures are still an open challenge.
- **Data Augmentation for Rare Pathologies**: Rare forms of pneumonia, such as those caused by uncommon pathogens or conditions like aspiration pneumonia, are often underrepresented in datasets. Current data augmentation strategies are insufficient to address this imbalance effectively.
- **Explainability and Trust in Clinical Settings**: While explainability is frequently discussed, there is a lack of practical, clinician-focused methods for interpreting model outputs in real-time. Models need to provide visual or textual explanations aligned with medical reasoning to gain trust in clinical practice.
- **Scalability in Resource-Limited Settings**: Although transformer-based models achieve high accuracy, their computational overhead limits scalability. Developing lightweight and energy-efficient architectures without compromising performance is essential for deployment in low-resource environments.
- **Integration of Segmentation and Classification**: Pneumonia detection pipelines often treat lung segmentation and pneumonia classification as independent steps. A unified model that jointly optimizes these tasks could improve overall accuracy and efficiency.

Addressing these gaps will enable the development of more generalizable, robust, and clinically meaningful solutions for pneumonia detection in real-world scenarios.

2 MATERIALS AND METHODS

2.1 Dataset

For segmentation, we have used two separate datasets, one for training and one for testing. The train dataset is "Chest X-ray Dataset for Tuberculosis Segmentation". There are two types of images: Tuberculosis positive and Normal. This is a fairly balanced dataset.

Table 1: Chest X-ray Dataset for Tuberculosis Segmentation

	Data
Total	704
Positive	345
Normal	359

For testing, the "Lung segmentation from Chest X-Ray dataset" was used.

Table 2: Lung segmentation from Chest X-Ray dataset

	Data
Total	138
Normal	80
Abnormal	58

For classification, we are using the Kermany dataset [1]. This dataset is divided into test and train and each has 2 classes: NORMAL and PNEUMONIA. The detailed statistics of this dataset are given below:

Table 3: Kermany Dataset Distribution

	Train	Test
Total	5232	624
Normal	1349	234
Pneumonia	3883	390

Here we can see that the training dataset is somewhat imbalanced, with nearly two-third of the data being PNEUMONIA classified. Test data is fairly balanced.

2.2 Proposed Architecture

2.2.1 Base Model. First we implemented the base model on our side. The input image is processed through a ResNet backbone, which consists of three layers: *Layer1*, *Layer2*, and *Layer3*. The outputs from these layers are passed through separate convolutional layers: *Conv_Red1*, *Conv_Red2*, and *Conv_Red3*, respectively. The outputs of *Conv_Red1* and *Conv_Red2* are concatenated (denoted as *Concat (L1 + L2)*), combining features from the first two layers into a single representation. Then two separate transformer modules are applied. *Transformer_Cat* processes the concatenated features from *L1* and *L2*, and *Transformer_3* processes the features from *Conv_Red3* (output of *Layer3*). The processed features are passed to the *Final Output* layer, which generates the model's predictions. In summary, this architecture combines the hierarchical feature extraction capabilities of ResNet with the global context modeling of transformers. By integrating outputs from multiple layers and applying separate transformers to different feature sets, the model effectively captures both local and global information. The concatenation and fully connected layers consolidate this information for the final prediction. [Figure 2]

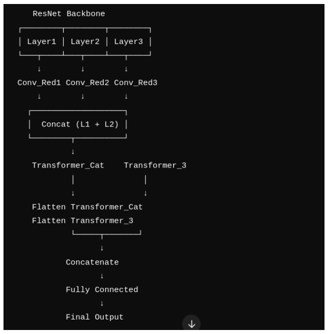

Figure 2: Base Model Architecture Overview

2.2.2 *Approach 1.*

The model architecture begins with a ResNet backbone, which extracts hierarchical features from the input image through three layers: *Layer1*, *Layer2*, and *Layer3*. Each layer outputs feature maps that are passed through separate convolutional layers named1*Conv_Red1*, *Conv_Red2*, and *Conv_Red3*, which reduce the dimensionality of these feature maps. The reduced features are then fed into three separate transformer modules: *Trans1* processes features from *Conv_Red1*, *Trans_Cat* processes concatenated features from *Conv_Red1* and *Conv_Red2*, and Trans3 processes features from *Conv_Red3*. The outputs of these transformers are then passed through fully connected layers (FC1, FC2, and FC3) to generate intermediate outputs. These intermediate outputs are combined using an attention-based dynamic weighting mechanism. The weighted sum of these outputs is finally passed through the Final Output layer, producing the model's prediction. This architecture effectively combines hierarchical feature extraction with transformer-based global context modeling, while leveraging dynamic weighting to adaptively fuse information from multiple pathways for robust prediction. [Figure 3]

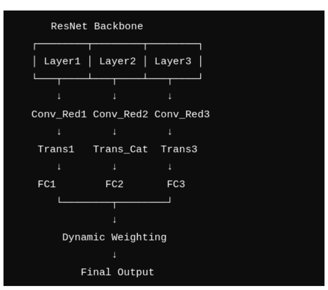

Figure 3: Approach 1 Architecture Overview

2.2.3 *Approach 2.*

This approach enhances the base model by introducing attention-weighted dynamic fusion and auxiliary losses to improve robustness and feature learning.

Attention-Weighted Dynamic Fusion: Instead of traditional dynamic weighting, this method employs an attention mechanism to adaptively assign weights to outputs from different layers. The attention mechanism dynamically evaluates the relevance of features extracted by *Transformer_1* and *Transformer_2*, ensuring that the model focuses on the most informative representations. This allows the model to emphasize critical features while suppressing less relevant ones, leading to improved performance.

Auxiliary Losses: To further enhance the learning process, auxiliary losses are introduced for the intermediate outputs from *Transformer_1* and *Transformer_2*. These additional losses are incorporated into the overall training objective, encouraging intermediate layers to learn meaningful and discriminative features. This not only helps stabilize training but also ensures that the intermediate layers contribute effectively to the final predictions.

By combining attention-weighted feature fusion with auxiliary losses, this approach improves the model's ability to adaptively focus on important features and learn robust representations, ultimately enhancing its effectiveness for pneumonia detection.

2.2.4 *Approach 3.*

In this approach, we extend the base model by integrating a Vision Transformer (ViT) backbone (`ViT_B_16`. The model extracts features from three intermediate layers (Layer 3, Layer 6, and Layer 11) of the ViT backbone. These layers capture different levels of abstraction, from low-level details to high-level semantic information.

Instead of directly concatenating the features, we introduced an attention gate mechanism to calculate the relative importance of each feature map. The attention gate assigns dynamic weights to the features extracted from each intermediate layer, ensuring that the model prioritizes the most informative representations.

Once the weights are calculated, the weighted features from the intermediate layers are concatenated to form a comprehensive feature vector. This combined representation is then passed through a fully connected layer to refine and prepare the features for the final classification.

By dynamically weighting and merging the hierarchical features from multiple layers, this approach enhances the model's ability to balance local and global context, leading to more robust and accurate predictions for pneumonia detection.

2.2.5 *Approach 4.*

In this approach, we extend the base model by incorporating both a ResNet backbone and a Vision Transformer (ViT) backbone. This hybrid design aims to leverage the strengths of both convolutional and transformer-based architectures.

First, features are extracted from three intermediate layers of the ResNet backbone. The features from the first two layers are concatenated, while the feature map from the third layer is kept separate. These processed feature outputs are denoted as `fg_embed_cat` and `fg_embed_3`, respectively.

Simultaneously, the ViT backbone processes the same input image, generating a global feature representation. The outputs from the ResNet and ViT backbones are then concatenated to form a comprehensive feature set.

To ensure an optimal combination of features, we employ a dynamic weighting mechanism. This mechanism assigns adaptive weights to the concatenated features before they are passed through a fully connected (FC) layer. The FC layer refines the weighted features and produces the final classification output.

By combining hierarchical feature extraction from ResNet and global context modeling from ViT, this approach balances local and global information, enhancing the model's robustness and performance for pneumonia detection. [Figure 4]

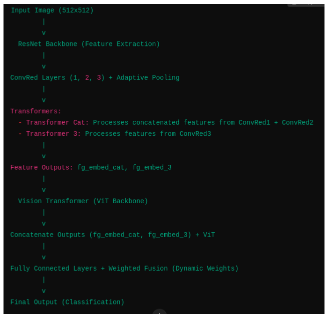

Figure 4: Approach 4 Architecture Overview

2.2.6 Approach 5. This approach is identical to **Approach 3**, with the only difference being the choice of backbone. Instead of using `ViT_B_16`, this approach employs `ViT_L_16` as the backbone.

By utilizing the larger `ViT_L_16` model, this approach aims to leverage its increased capacity for capturing richer global contextual information from the input images. The rest of the methodology, including the extraction of hierarchical features, attention-based dynamic weighting, and feature concatenation, remains unchanged.

The use of `ViT_L_16` provides an opportunity to evaluate the impact of a larger transformer backbone on the overall model performance for pneumonia detection.

2.3 Training Details

2.3.1 Loss Functions. The loss functions used in our framework are designed to address the specific requirements of segmentation and classification tasks effectively. Below, we outline the loss functions in detail:

- Segmentation Loss: Combined BCE and Dice Loss
  - We use a combined loss function that integrates Binary Cross-Entropy (BCE) Loss and Dice Loss.
  - Binary Cross-Entropy (BCE) Loss: This loss measures the pixel-wise classification error for segmentation masks. It is defined as:

$$\text{BCE Loss} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$$

    where:
    - ∗ N is the total number of pixels.
    - ∗ y_i is the true label for pixel i (1 for foreground, 0 for background).
    - ∗ p_i is the predicted probability for pixel i.
  - Dice Loss: This loss measures the overlap between predicted and ground truth masks. It is particularly useful for imbalanced datasets. Dice Loss is defined as:

$$\text{Dice Loss} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon}$$

    where:
    - ∗ y_i is the ground truth value for pixel i.
    - ∗ $\hat{y}_i$ is the predicted value for pixel i.
    - ∗ ϵ is a small constant added for numerical stability.
  - Combined Loss: The two components are combined with a weighting factor to balance their contributions:

$$\text{Combined Loss} = \alpha \cdot \text{BCE Loss} + \beta \cdot \text{Dice Loss}$$

    where α and β are weights that balance the contributions of the two losses.
- **Classification Loss: Cross-Entropy Loss**
  - For pneumonia classification, we use Cross-Entropy Loss, which is a standard choice for binary or multi-class classification tasks.
  - It penalizes incorrect predictions by comparing the predicted probability distribution with the true labels. The Cross-Entropy Loss is defined as:

$$\text{Cross-Entropy Loss} = -\frac{1}{M}\sum_{j=1}^{M}\sum_{k=1}^{C} y_{j,k}\log(p_{j,k})$$

    where:
    - ∗ M is the total number of samples.
    - ∗ C is the number of classes.
    - ∗ $y_{j,k}$ is a binary indicator (1 if class k is the correct label for sample j, 0 otherwise).
    - ∗ $p_{j,k}$ is the predicted probability of sample j belonging to class k.

These loss functions ensure precise pixel-level segmentation, effective handling of class imbalance, and robust classification of pneumonia cases.

3 RESULTS AND DISCUSSION

3.1 Results

3.1.1 Base Model. We attempted to replicate the results of the base model. However, due to resource limitations and probable difference in data preprocessing we were unable to achieve those results. We ran this model for 20 to 43 epochs and achieved the results in table 4. If we look at the following figure, we see that while training loss is somewhat stabilizing, validation loss keeps spiking. The loss has not stabilized after running for 43 epochs.

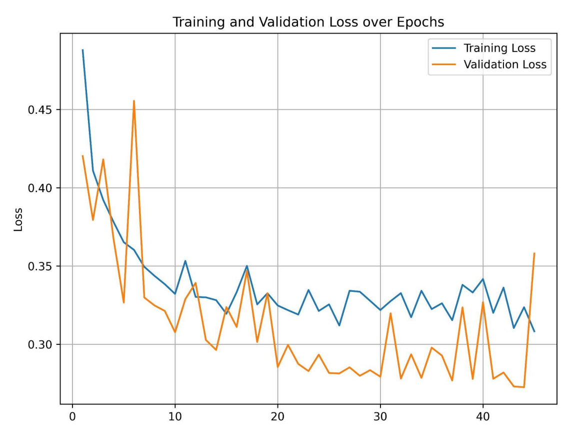

Figure 5: Training and Validation Loss over 43 Epochs

3.1.2 Approach 1. The four transformer approach, where features have been extracted from the intermediate layers of the resnet backbone achieves following results after 40 epochs. Validation loss remains highly unstable and is now greater than training loss. Refer to table 5 for performance metrics details.

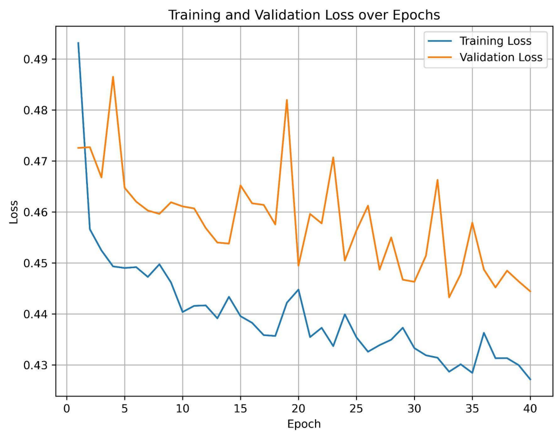

Figure 6: Training and Validation Loss over 40 Epochs

3.1.3 Approach 2. The auxiliary approach, achieves following results after 20 epochs. Both train and validation loss have stabilized so we stopped at 20 epochs. Here train loss is much higher than vaalidation loss as train loss incorporates intermediate losses. Refer to table 6 for performance metrics details.

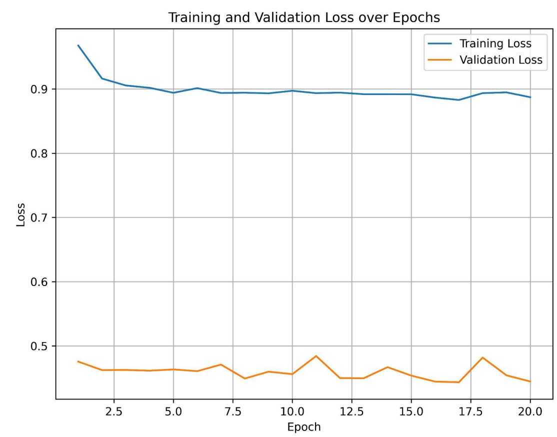

Figure 7: Training and Validation Loss over 20 Epochs

3.1.4 Approach 3. This approach has performed the worst among all 5. Although train loss smoothly stabilized, validation loss kept increasing over 30 epochs. Performance metrics details are shown in table 7.

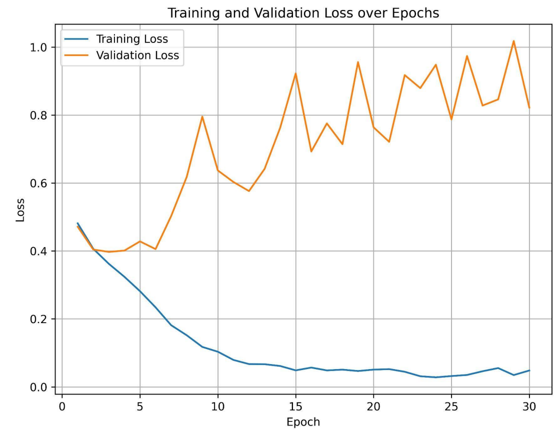

Figure 8: Approach 3: Training and Validation Loss over 30 Epochs

3.1.5 Approach 4. In this approach validation loss kept having high spikes over 25 epochs. Performance metrics details are shown in table 8.

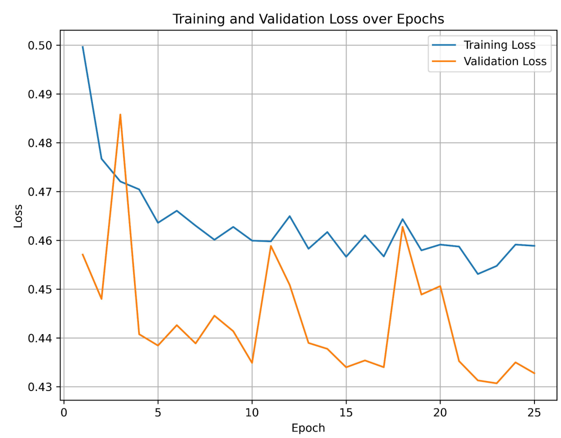

Figure 9: Approach 4: Training and Validation Loss over 25 Epochs

3.1.6 Approach 5. Performance metrics details shown in table 9.

Table 4: Performance Metrics Over Epochs: Base Model

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
20	81.09%	0.84	0.76	0.78	0.76	0.90	0.93
25	81.73%	0.84	0.77	0.79	0.77	0.90	0.93
30	81.41%	0.84	0.77	0.78	0.77	0.90	0.93
35	83.01%	0.83	0.80	0.81	0.80	0.90	0.93
40	74.36%	0.83	0.66	0.66	0.66	0.90	0.93
43	81.57%	0.83	0.77	0.79	0.77	0.90	0.93

Table 5: Performance Metrics Over Epochs: Approach 1

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
20	79.01%	0.79	0.76	0.77	0.76	0.87	0.92
25	79.33%	0.78	0.78	0.78	0.78	0.86	0.91
30	78.21%	0.78	0.74	0.75	0.74	0.86	0.91
35	75.16%	0.80	0.68	0.69	0.68	0.86	0.91
40	77.88%	0.77	0.75	0.76	0.75	0.86	0.91

Table 6: Performance Metrics Over Epochs: Approach 2

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
20	78.21%	0.77	0.76	0.76	0.76	0.87	0.92

Table 7: Performance Metrics Over Epochs: Approach 3

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
20	74.84%	0.73	0.73	0.73	0.73	0.80	0.85
25	71.96%	0.70	0.68	0.68	0.68	0.77	0.84
30	75.32%	0.74	0.72	0.73	0.72	0.80	0.85

Table 8: Performance Metrics Over Epochs: Approach 4

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
10	77.08%	0.81	0.71	0.72	0.71	0.87	0.92
15	79.49%	0.79	0.76	0.77	0.76	0.87	0.92
25	79.65%	0.79	0.76	0.77	0.76	0.87	0.92

Table 9: Performance Metrics Across Epochs: Approach 5

Epochs	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test Specificity	Test AUC	Test AUPRC
10	72.76%	0.75	0.66	0.66	0.66	0.72	0.73
20	73.56%	0.73	0.69	0.69	0.69	0.73	0.78

3.2 Discussion

In our project we have experimented with various architectural changes in order to improve the base model. Due to potential data preprocessing issues and resource limitations, we were unable to replicate or improve the results from the base model. We have explored the integration of convolutional backbones and transformer-based architectures to improve pneumonia detection in pediatric chest X-rays. By using ResNet and Vision Transformer (ViT) backbones alongside position-aware embeddings and transformer modules, the proposed framework strived to achieve enhanced classification accuracy and robust performance metrics. The dynamic weighting mechanism introduced allows for effective fusion of features from different architectural modules, ensuring that the final predictions leverage both local and global feature representations. The comparative analysis of multiple architectural approaches demonstrates that combining transformers and convolutional performs better compared to standalone models. However, the results also reveal challenges with balancing computational complexity and performance. Specifically, the addition of ViT to a ResNet backbone with multiple transformers did not always yield expected improvements, suggesting potential redundancies or overfitting when combining architectures.

It's also worth noting that these experiments underline the importance of carefully tuned hyperparameters, balanced datasets, and preprocessing techniques in achieving high performance. However, the issue of resource limitation needs to be tackled in future works.

4 CONCLUSION

This project explores an innovative approach to pneumonia detection using multi-scale transformer models integrated with advanced vision backbones. By leveraging a combination of a ResNet backbone and transformer-based modules, as well as dynamic weighting mechanisms, the framework effectively captures hierarchical features and global contextual information. Additionally, the integration of the ViT model in certain approaches demonstrated its potential in extracting more granular features from input data. The experiments reveal that the dynamic weighting mechanism enables adaptive fusion of information, improving the robustness of predictions in many scenarios.

However, the results also highlighted some limitations. Certain architectural combinations, such as excessive stacking of models, led to overfitting or degraded validation performance, as observed in some training loss spikes. Resource constraints and dataset preprocessing also played a role in limiting model performance compared to baseline benchmarks.

For future work, the focus could shift to optimizing the dynamic weighting mechanism further, introducing auxiliary regularization techniques to enhance stability, and experimenting with lightweight transformer architectures to reduce computational overhead. Working with better resources would also be recommended. Additionally, expanding the dataset and incorporating domain adaptation techniques could improve the model's generalizability to other medical imaging tasks.

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